



**Department of Electrical and
Computer Engineering**

SENG 426 - B01

Lab Project 3

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I. Test Cases with Test Data Value

Test Case	Test Data	Expected Result
shouldLoginSuccessfully	ADMIN_EMAIL, ADMIN_PASSWORD from testing/.env	Admin logs in and sees Welcome back, Admin!
shouldFailLoginWithInvalidC redentials	invalid@vega.com, invalidpassword	Login fails with credential error
shouldRejectSqlInjectionAtte mpt	Email fakeemail@vega.com, password x' OR '1' = '1	Login is rejected
checkThatSignUpScenarioW orkingSuccessfully	Random name/email, 6-digit password, role User	Account registers and waits for admin approval
pendingUserCannotLoginBe foreAdminApproval	Same newly registered user before activation	Login is blocked pending approval
approveRegistrationRequest	Admin credentials + newly registered user	Admin activates account successfully

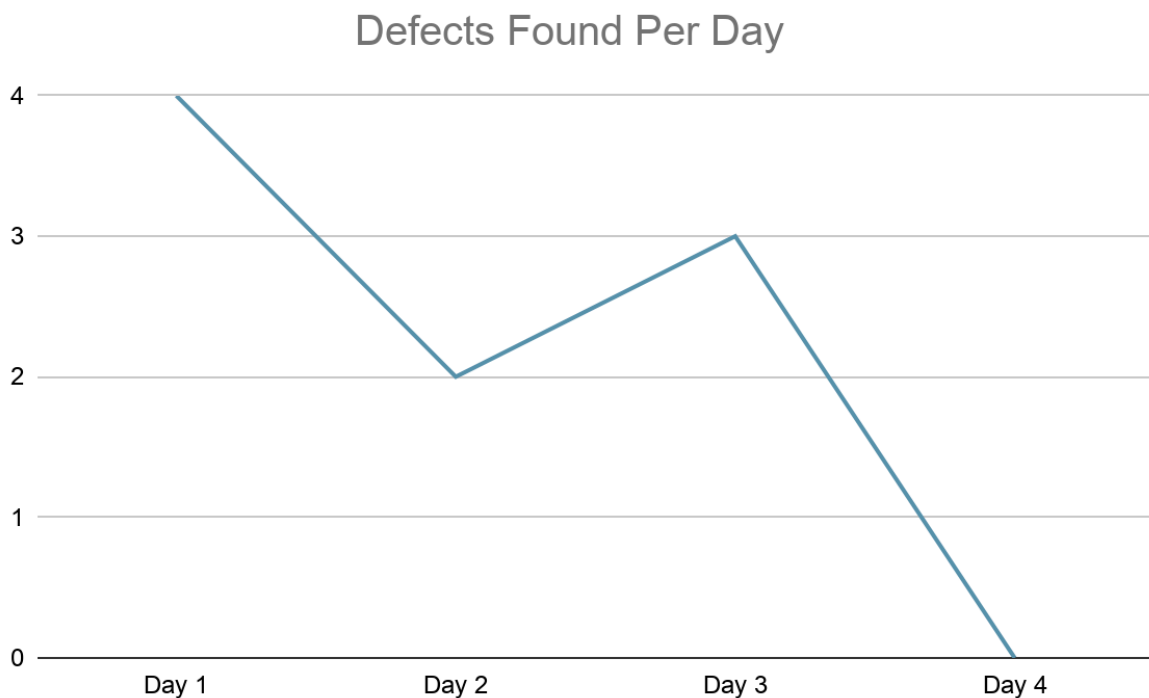
adminCanUploadResourceFilesSuccessfully	Admin credentials + temporary .txt file	File uploads successfully
standardUserCannotAccessResourceByUrl	LocalStorage user with role USER	/resources access is denied
adminCanAccessDefenseDashboard	Admin credentials	Defense dashboard opens for admin
unauthenticatedUserCannotAccessDefenseDashboardByUrl	No login/session	Direct dashboard access redirects to Login
publicRegistrationShouldRejectAdminRole	API registration body with role ADMIN	Request is rejected with 4xx
roleChangeWithoutAdminAuthenticationShouldBeRejected	API role-change request without token	Request is rejected with 401/403

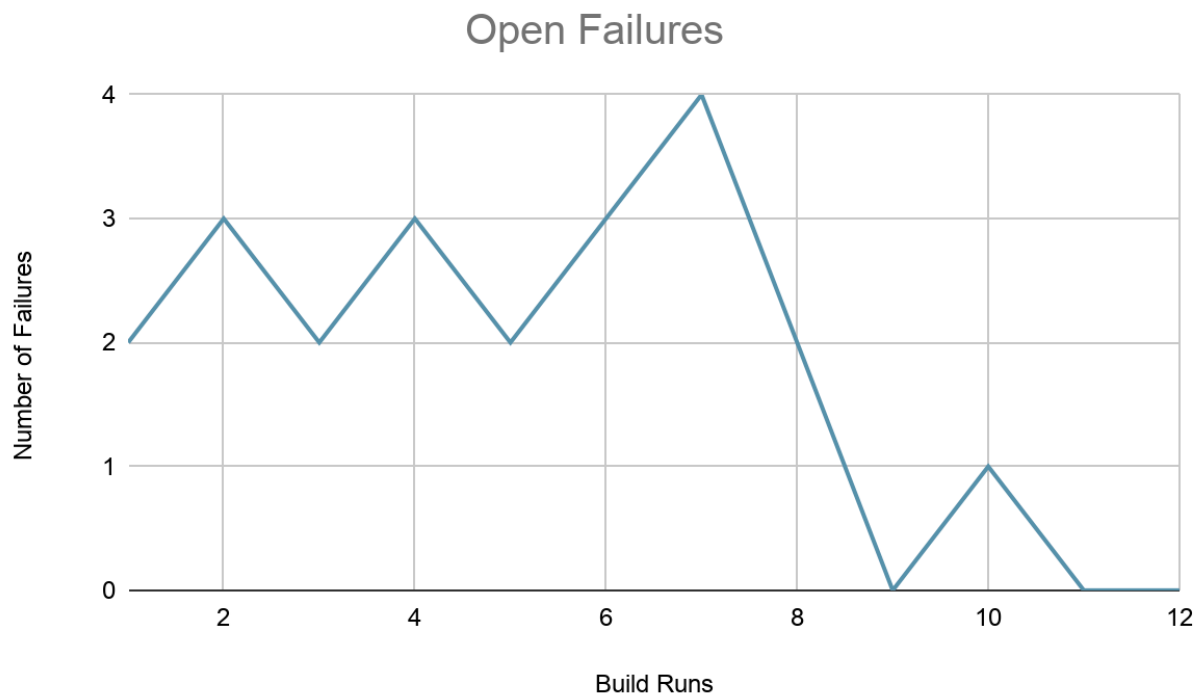
II. Bug and Pull Request Descriptions

Issue	Bug Description	Severity	Status
#15	SQL Injection in User Login Repository: Unparameterized SQL concatenation in CustomUserLoginRepository allowed authentication bypass via malicious password payloads.	Critical	Fixed
#20	Employee Resources Route Exposure: Standard Users could access employee-restricted company files by manually navigating to /resources.	High	Fixed
#9	HTTP 415 Media Type Upload Rejection: System rejected valid resource file uploads under 50 MB due to incorrectly formatted multipart FormData construction.	High	Fixed
#23	Public Registration can Request Admin Role: Public registration API accepts ADMIN as a role value. The frontend hides this option; however, they are still exposed because of the backend API schema and role endpoints.	High	Fixed
Original login failure	Valid admin login test was failing before login flow/security fixes.	High	Fixed
Original registration approval failure	Admin approval workflow test was unstable and failing.	High	Fixed
Task 1 coverage gap	Defense Dashboard access needed coverage.	Medium	Fixed

Pull Request Title	Description
Fixes for Issues 9/20	Fix for issues 9 and 20, as they are close in scope, work was done to resolve the two original tests that were failing too.
Resolve "Public registration can request ADMIN role"	Fixes issue 23, where users could request access for the ADMIN role.
Validate login by password then email	Fixes issue 15, where the login endpoint builds a native SQL query that concatenates user-controlled email and password values. Comparisons for credentials are done against plaintext password values, instead of using a configured password encoder.
Improved testing coverage for task 1, addition of a readme file	Fixes added for the Defense Dashboard and Admin permissions along with readme file including instructions used to replicate and test bug fixes.

III. Build Statistics Graphs





IV. Defect Dynamic Model and DRE Measure

Phase	Defects	Defects Fixed	DRE (%)
Initial Automated Testing	2	2	100
Issue-specific testing	4	4	100%
Merge/regression testing	1	1	100%
Final regression suite	0	0	N/A
Overall	7	7	100%

V. Commentary on Quality Characteristic

Before the task, the app had critical security vulnerabilities. Post-task, with automated tests and fixed bugs, the app's reliability and security have significantly improved, and tests no longer carry stale states between runs, ensuring test results are more accurate.

Completion of the test creation task also increased the quality of the code we wrote after it, as test-driven development ensured that the fixes we implemented for these issues in task two were aligned with the expectations and requirements outlined for the software.

The fixes themselves elevate our security. Removing possible attack vectors such as SQL injection provided the opportunity to investigate the login code and modify it to encode new passwords and old passwords as users login following the update. The fixes also help to protect against unauthorized access or elevation of privilege, and the automated testing pipeline ensures that no future changes will reintroduce these issues with future updates. Our changes changed our app from using unverified assumptions about security to explicit assertions for security and authorization.

VI. Screenshots of Pipeline Build History, CI/CD Analytics Graph, and Test Result Reports

All32

Finished

Branches

Tags

View analytics

Clear runner caches

New pipeline

Filter pipelines

Q

Show Pipeline ID ▾

Status	Pipeline	Created by	Stages	Actions
✓ Passed 00:04:13 3 minutes ago	#7062 Further fixes to issue 15 fix, syntax fix, is... 15 development 3abda3f4 latest branch		✓	↓ ▾
✗ Failed 00:01:35 32 minutes ago	#7060 Merge branch '15-user-login' into 'develo... 15 development 104651d4 branch		✗	↺ ↓ ▾
✗ Failed 00:00:13 34 minutes ago	#7059 Merge branch 'development' into '15-use... 15 15-user-Login dfd0bbfa latest branch		✗	↺ ↓ ▾
✗ Failed 00:00:15 45 minutes ago	#7057 added sql injection test 15 15-user-Login c0bef602 branch		✗	↺ ↓ ▾
✗ Failed 00:00:28 1 hour ago	#7049 password encoding at creation 15 15-user-Login 934ab15f branch		✗	↺ ↓ ▾
✓ Passed	#7044 One of the tests was still failing since the		✓	↓ ▾

Figure 1: Pipeline Build History

CI/CD Analytics

Pipelines ?

Total pipeline runs
32

Failure rate
72%

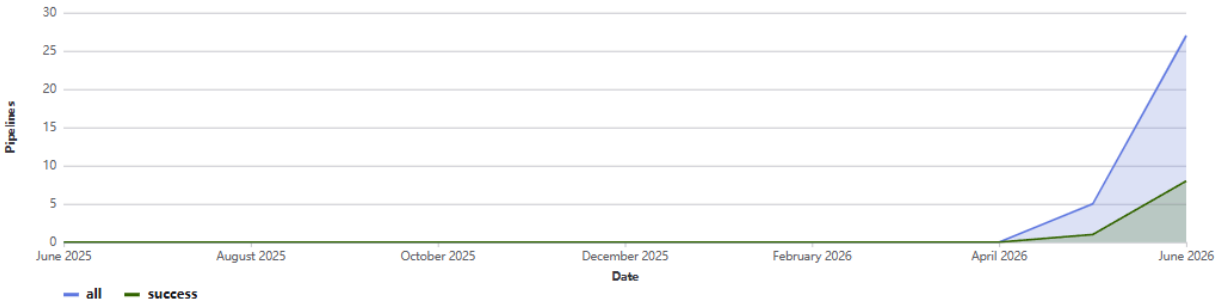
Success rate
28%

[View all](#)

Pipelines charts

Last week Last month Last year

Date range: Jun 24, 2025~Jun 24, 2026



Pipeline durations for the last 30 commits

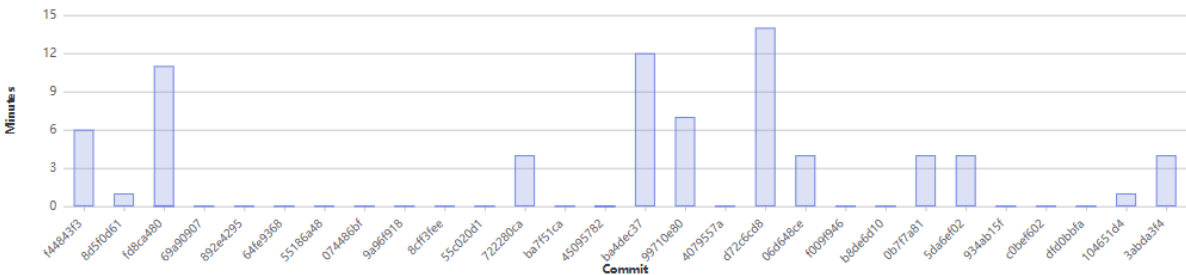


Figure 2: CI/CD Analytics Graph Throughout Development

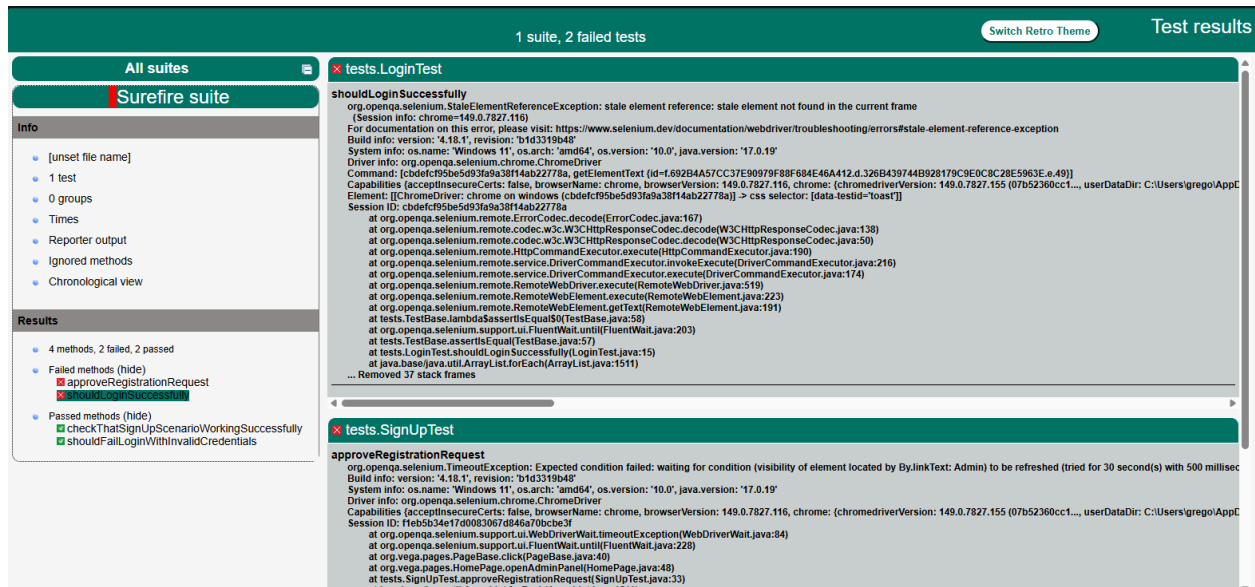


Figure 3: Initial build

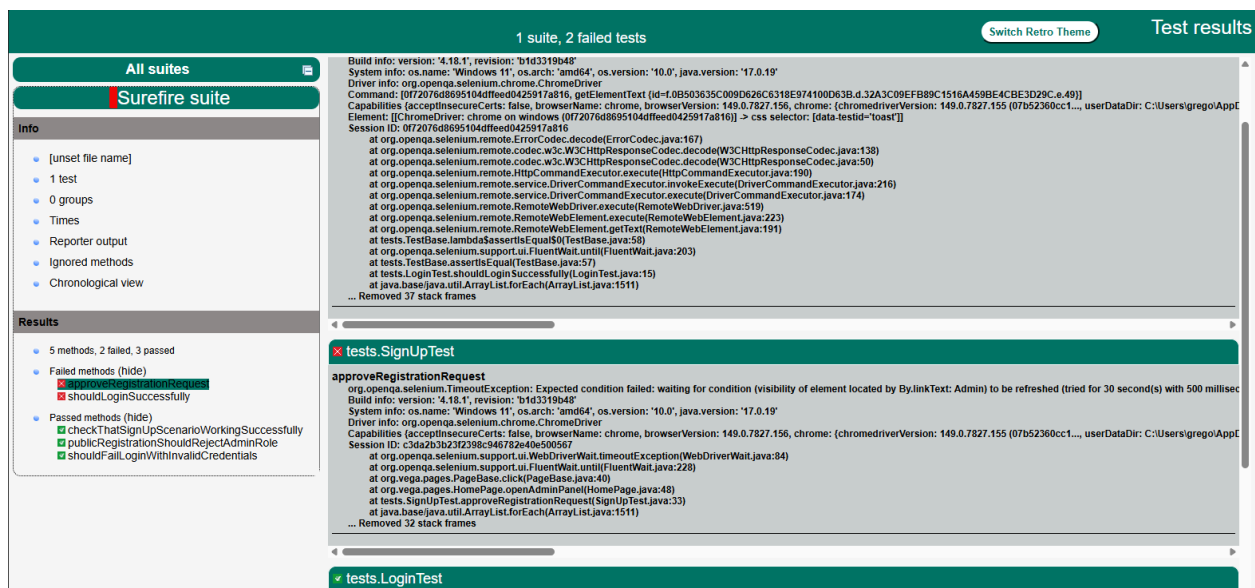


Figure 4: Build with Issue 23 Fixed

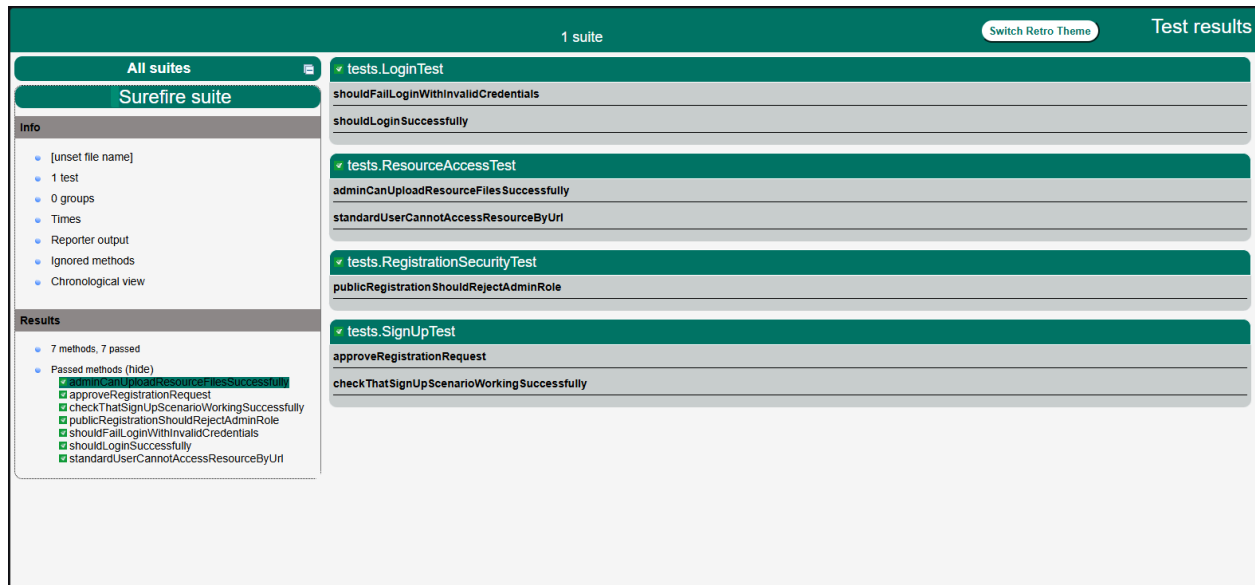


Figure 5: Build with Issues 9 and 20 Fixed

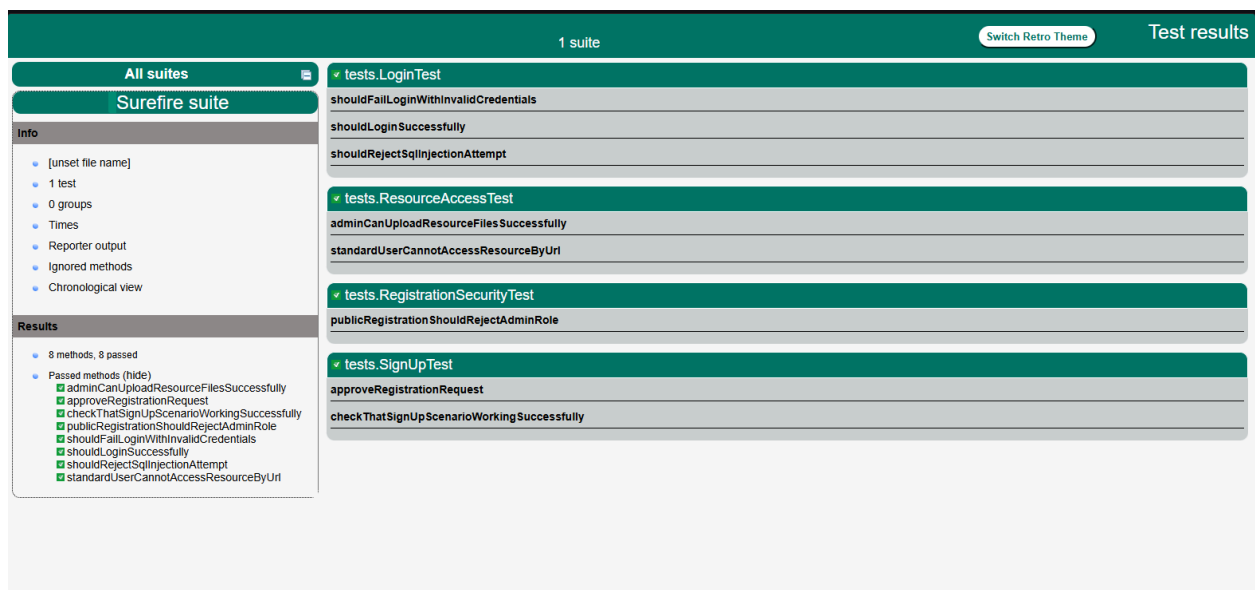


Figure 6: Build with Issue 15 Fixed

1 suite		Switch Retro Theme	Test results
All suites			
Surefire suite			
Info			
[unset file name] 1 test 0 groups - Times - Reporter output - Ignored methods - Chronological view			
Results			
10 methods, 10 passed - Passed methods (hide) - adminCanUploadResourceFilesSuccessfully - approveRegistrationRequest - checkThatSignUpScenarioWorkingSuccessfully - pendingUserCannotLoginBeforeAdminApproval - publicRegistrationShouldRejectAdminRole - roleChangeWithoutAdminAuthenticationShouldBeRejected - shouldFailLoginWithInvalidCredentials - shouldLoginSuccessfully - shouldRejectSqlInjectionAttempt - standardUserCannotAccessResourceByUrl			
		tests.LoginTest shouldFailLoginWithInvalidCredentials shouldLoginSuccessfully shouldRejectSqlInjectionAttempt	
		tests.ResourceAccessTest adminCanUploadResourceFiles Successfully standardUserCannotAccessResourceByUrl	
		tests.RegistrationSecurityTest publicRegistrationShouldRejectAdminRole roleChangeWithoutAdminAuthenticationShouldBeRejected	
		tests.SignUpTest approveRegistrationRequest checkThatSignUpScenarioWorkingSuccessfully pendingUserCannotLoginBeforeAdminApproval	

Figure 7: Build with added Admin Test Coverage

1 suite		Switch Retro Theme	Test results
All suites			
Surefire suite			
Info			
[unset file name] 1 test 0 groups - Times - Reporter output - Ignored methods - Chronological view			
Results			
12 methods, 12 passed - Passed methods (hide) - adminCanAccessDefenseDashboard - adminCanUploadResourceFilesSuccessfully - approveRegistrationRequest - checkThatSignUpScenarioWorkingSuccessfully - pendingUserCannotLoginBeforeAdminApproval - publicRegistrationShouldRejectAdminRole - roleChangeWithoutAdminAuthenticationShouldBeRejected - shouldFailLoginWithInvalidCredentials - shouldLoginSuccessfully - shouldRejectSqlInjectionAttempt - standardUserCannotAccessResourceByUrl - unauthenticatedUserCannotAccessDefenseDashboardByUrl			
		tests.DefenseAppAccessTest adminCanAccessDefenseDashboard unauthenticatedUserCannotAccessDefenseDashboardByUrl	
		tests.RegistrationSecurityTest publicRegistrationShouldRejectAdminRole roleChangeWithoutAdminAuthenticationShouldBeRejected	
		tests.LoginTest shouldFailLoginWithInvalidCredentials shouldLoginSuccessfully shouldRejectSqlInjectionAttempt	
		tests.ResourceAccessTest adminCanUploadResourceFiles Successfully standardUserCannotAccessResourceByUrl	
		tests.SignUpTest approveRegistrationRequest checkThatSignUpScenarioWorkingSuccessfully pendingUserCannotLoginBeforeAdminApproval	

Figure 8: Final Build

Test Execution History Table

Run	Tests Run	Passed	Failed	Notes
1	4	2	2	Initial Selenium suite had failures in original login and registration approval tests
2	5	2	3	Issue #23 test added, admin role registration security test failed
3	5	3	2	Issue #23 resolved
4	6	3	3	Issue #9 test added, file upload test failed
5	6	4	2	Issue #9 test passed after the upload/FormData fix
6	7	4	3	Issue #20 test added to check resource access
7	7	3	4	Issue #9 and #20 failed during integration
8	7	5	2	Issues #9 and #20 resolved
9	7	7	0	Original approveRegistrationRequest and shouldLoginSuccessfully failures fixed
10	7	6	1	Merge reintroduced issue #20 failure
11	7	7	0	Upload path and resource resolution fixed
12	12	12	0	Final expanded Task 1 suite implemented and passed

VII. Table with Titles, Original Estimation, and Actual Time Spent on Each Tick

Ticket Title	Original Estimation (hours)	Actual Time Spent (hours)
Public registration can request admin role	5	6
File Upload Failure	3	2
User webpage bypass through URL	6	8
User Login	5	5